AMRO AL-BAALI

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SKILLS

Domain Knowledge: Sensor fusion, state estimation, localization, SLAM, pose graph optimization, computer vision, multi-sensor extrinsic calibration, matrix Lie groups, linear algebra.
Programming Languages: C++, Python, MATLAB.
Libraries and Tools: ROS, Eigen, Numpy, OpenCV, Ceres, GTSAM, NvBlox.

EDUCATION

M. Eng Mechanical **McGill University**

05/2019 - 08/2021 Montreal, Canada

- CGPA: 3.77/4.00. Supervisor: Prof. J. R. Forbes.
- Thesis title: *Augmenting Sensor Measurements with INS Estimates in a Graph Based SLAM Problem*.
- Publication: A. Al-Baali, T. Hitchcox and J. R. Forbes, "Combining DVL-INS and Laser-Based Loop Closures in a Batch Estimation Framework for Underwater Positioning," in *IEEE Journal of Oceanic Engineering*, doi: `10.1109/JOE.2023.3286854`.

B. Eng Honours Mechanical, Minor in Computer Science **McGill University**

09/2014 - 04/2019 Montreal, Canada

- CGPA: 3.83/4.00. Supervisor: Prof. J. R. Forbes.
- Thesis title: *Parallel Feedforward Control Using Linear Matrix Inequalities*.

WORK EXPERIENCE

Senior SLAM Engineer & Team Lead - Scene Perception **Pickle Robot**

02/2024 – Present Cambridge, MA

- Designed and implemented the real-time obstacles (3D) mapping stack that generated ESDFs used by Motion Planning and control to generate and execute collision-free arm trajectories. The stack fused 3D lidars and package perception estimates leveraging libraries such as NvBlox, cuRobo (mapper), and KISS-ICP, generating a dense ESDF for consumption by the Motion Planning team.
- Designed and prototyped a real-time detection and tracking system that estimates the 3D pose of telescopic conveyors with respect to the AMR's conveyor, enabling the navigation subsystem to align the robot with the telescopic conveyor.
- Led the design, development, and production deployment of a complete localization and semantic mapping system shipped to customer sites, fusing 2D lidar odometry with semantic mapping of container walls and ramps, robust against wheel slip while the robot arm picks packages.
- Tech lead for the Scene Perception team (4–5 engineers), driving the design and development of multi-modal sensor extrinsic calibration, scene perception, and obstacles mapping.
- Built a sensor data playback pipeline decoupling perception algorithm development from hardware, enabling rapid, reproducible iteration on localization and state estimation, which reduced the need to run on-robot tests, which in turn improved the team's velocity.

Robotics Software Engineer - Robotics Team **Vention**

05/2023 – 01/2024 Montreal, Canada

- Designed and implemented a new feature that brought a new type of user (power users) interested in directly controlling the robot through programming it. I worked on the architecture that allows the user to simulate the commands before executing them on the robot, allowing for faster prototyping for the customer.
- Worked closely with customers to introduce the new feature to them and customize it for their needs. I provided periodic updates including prototypes the customers can test on.
- Improved the process of prototyping on **ROS**, which was used to run the robot simulations, by automating the build and deployment processes. My changes reduced friction in the development process, allowing developers to prototype more quickly and to introduce fewer bugs.

Software Engineer - Localization and Mapping **Avidbots**
09/2021 – 03/2023 Kitchener, Canada

- Developed and maintained the lidar-based localization and mapping pipeline, directly improving the runtime efficiency of the least-squares scan-matcher used to localize the robot against a pre-existing map.
- Designed and shipped new localization features, including localization close to walls (enabling wall following), and visual-based localization for improved initial-localization.
- Led the effort of adapting the wheel odometry pipeline to accommodate the removal of wheel encoders to reduce the robot cost. This involved developing a robust online steering-wheel encoders calibration routine and changing the kinematic model.
- Led the effort of setting up an OptiTrack system to evaluate the localization performance during in-house tests. The effort included the design and implementation of a calibration routine that aligned the OptiTrack coordinate system with the map used in the localization pipeline.

Graduate Student - SLAM **DECAR group (McGill University)**
05/2019 – 08/2021 Montreal, Canada

- Collaborated with Voyis and Sonardyne to develop a SLAM algorithm for an Autonomous Underwater Vehicle (AUV) equipped using a third-party Inertial Navigation System (INS) and the Voyis Insight Pro high-precision laser scanner.
- Researched and designed a novel localization protocol that treats the INS as a black box and fuses its output with Voyis' high-precision laser scanner, which was used to generate highly precise maps of the seabed used by Voyis for various applications. The algorithm leveraged convex and on-manifold optimization.
- Built an on-manifold factor graph solver in MATLAB, used as the backend for the AUV trajectory optimization, jointly estimating pose and landmark states over $SE(3)$ manifolds via nonlinear least-squares optimization.

Mechanical Engineering Intern **MY01**
05/2018 – 04/2019 Montreal, Canada

- Designed and executed mechanical tests on the MY01 device to obtain medical certification, such as stress tests of the needle-penetration mechanism and plastic enclosure.
- Designed, implemented, and tuned a PID controller to precisely regulate the speed and force of a Mark10 force-gauge rig, enabling repeatable force- and speed-controlled penetration tests.
- Built the full application stack for the test rig from scratch, including a GUI for operators to configure and run controlled experiments and real-time data visualization to support certification reporting.
- Customized the Autodesk Vault software tool used for CAD version control to automatically generate reports and maintain version numbers for released designs.

Undergraduate Research Assistant **McGill University**
05/2017 – 08/2017 Montreal, Canada

- Researched the use of multi-objective optimization methods, such as genetic algorithms, for design-space exploration (DSE), under the supervision of Prof. D. Varro.
- Used OpenMDAO and Platypus packages to implement the optimization for the DSE.

Teaching Assistant **McGill University**
09/2017 – 04/2021 Montreal, Canada

I was a teaching assistant for the following courses, providing tutorials, grading assignments, mid-terms and quizzes, and assisting students in understanding the course materials and assignments.

- MECH 513 (Control Systems) (Winter 2021)
- MECH 309 (Numerical Methods) (Fall 2019)
- MECH 412 (System Dynamics and Control) (Fall 2017)